



What is Footprinting?

Footprinting is the **first phase** of ethical hacking and penetration testing.

It involves **gathering information about a target system or organization** to find ways to infiltrate it.

Think of it as a hacker's research phase — like doing homework before an attack. Footprinting helps to:

- Understand the target's structure
- Identify weak points (like exposed IPs or domains)
- Plan attacks more efficiently



Types of Footprinting

There are **two main types** of footprinting:

◆ 1. Passive Footprinting

- **Definition:** Collecting information **without directly interacting** with the target.
- **Goal:** Remain undetected while gathering data.
- **Examples:**
 - Searching company details on Google
 - Looking up domain info via **Whois**
 - Finding employee names on LinkedIn
 - Checking websites using tools like **theHarvester**



Used during **Open Source Intelligence (OSINT)** activities.

◆ 2. Active Footprinting

- **Definition:** Collecting data by **directly interacting** with the target's system.
- **Goal:** Get more accurate and deep technical information.
- **Examples:**
 - Ping sweep or port scan
 - Using **Nslookup** to query DNS servers
 - Traceroute to discover network paths



More risky, because it may alert the target's firewall or IDS (Intrusion Detection System).

Common Footprinting Tools

Tool	Description	Type
Whois	Retrieves domain ownership, registration, email, and address	Passive
Nslookup	Queries DNS records (A, MX, NS, etc.)	Active
Recon-ng	Powerful OSINT tool in Kali Linux; modular framework	Passive
theHarvester	Collects emails, hosts, and IPs from public sources	Passive
Nmap	Port scanner, also gives host and service info	Active
Shodan	Search engine for internet-connected devices (IoT, servers)	Passive

Real-World Example

Let's say a hacker wants to attack www.company.com. Here's how footprinting would work:

Passive:

- Use Whois to find domain registration email and IP range.
- Google search to find exposed subdomains or admin login portals.
- Check job postings for tech stack (e.g., "hiring Python devs").

Active:

- Use Nslookup to resolve IP address of subdomains.
- Perform ping sweeps to find live hosts.
- Use Nmap to scan open ports on the server.

Summary Table

Aspect	Passive Footprinting	Active Footprinting
Visibility	Hidden	Detectable by target
Risk	Low	Medium to High
Data Accuracy	Limited	High
Examples	Whois, Google, LinkedIn Nslookup, Ping, Nmap	